

II. Related Work

A. Reasoning-Based Autonomous Driving and CoC

Connecting visual observations to language-based reasoning and actions is an important direction in autonomous driving. Alpmayo-R1 combines Chain of Causation (CoC) reasoning with trajectory prediction and uses supervised fine-tuning and reinforcement learning to improve reasoning-action alignment. DriveLM represents relationships among perception, prediction, and planning questions and answers as a graph, while DriveVLM connects scene description, scene analysis, and hierarchical planning. These studies provide learnable representations of the process from observation to decision. [1](#), [2](#), [3](#)

Alignment between explanations and actions is also important for useful reasoning. Drive-R1 addresses reasoning-planning misalignment and history-based shortcut learning, using trajectory and meta-action rewards after supervised fine-tuning. Complementing this direction, we focus on how behavioral constraints are represented in learning data. We distinguish CoC explanations of why an action is taken from constraints restricting which actions are admissible in the situation, and combine the two. This distinction defines explicit normative information to accompany explanations, rather than presuming a general performance deficiency of existing reasoning models. [4](#)

B. Traffic-Rule Specification and Regulation-Aware Decisions

Machine-interpretable traffic rules provide a direct foundation for formalizing behavioral constraints. Esterle et al. express traffic rules in linear temporal logic (LTL) and apply them to compliance assessment on driving data. This approach gives applicability conditions and temporal relationships explicit semantics. Following this principle, we structure the subject and target, activation, persistence, release, and evidence links of CoC-augmenting constraints as EBLC elements. [5](#)

Retrieval-augmented reasoning offers another way to connect regulations with scenes. DriveReg retrieves situation-relevant traffic regulations and connects them to language-model reasoning to support compliance and interpretable decisions. Its central focus is retrieving and using applicable regulations. GuardSynth instead focuses on expressing selected behavioral constraints in an explicit intermediate representation, checking them within a restricted logical model, and conveying them as natural-language learning inputs. Regulation retrieval and constraint specification have complementary roles in obtaining supporting evidence and representing constraints. [6](#)

C. Natural Language, Formal Specifications, and CNL

Translating natural-language requirements into formal specifications connects human-readable descriptions with machine-verifiable representations. NL2TL uses abstracted atomic propositions and language models to translate natural language into temporal logic, separating domain-specific proposition recognition from translation structure. nl2spec associates language fragments with subformulas and lets users revise these associations to resolve ambiguity. The former emphasizes learning and adapting a trans-

lation model, while the latter emphasizes interactive specification and user correction. 7, 8

Controlled Natural Language (CNL) is also relevant when specifications are presented in readable language. Attempto Controlled English (ACE) restricts grammar and interpretation rules to connect natural-language sentences with formal representations. GuardSynth’s CNL is not an implementation of ACE; it maps fields and clauses of a selected EBLC profile to restricted sentence patterns. Our focus is traceable correspondence between defined constraint elements and generated sentences, rather than semantic equivalence across unrestricted natural language. Logical verification, sentence correspondence, and the validity of scene evidence are consequently distinct review targets. 9

D. Positioning of This Work

These research directions establish foundations for learning driving reasoning, formalizing traffic rules, and connecting language with logic. Building on them, GuardSynth defines the scope, properties, and elements of CoC behavioral constraints and connects EBLC specification, restricted logical verification, CNL generation, and CoC augmentation. The focus is on organizing learning constraints through a consistent representation and verification procedure, rather than developing a new driving model or a general-purpose translator. Table 1 compares the central objectives of the selected studies; it is neither an exhaustive feature audit nor a performance ranking.

Table 1. Central objectives of related work and their relationship to GuardSynth.

Study	Central objective	Representation or approach	Connection to this work
Alpamayo-R1; Drive-R1 [1, 4]	Reasoning-action alignment	CoC/CoT, supervised learning and RL	Useful supervision for action selection
DriveLM; DriveVLM [2, 3]	Reasoning from scenes to plans	Graph question answering; hierarchical planning	Structuring scene and action explanations
Esterle et al. [5]	Machine-interpretable traffic rules	LTL specifications	Explicit conditional and temporal semantics
DriveReg [6]	Regulation-aware decisions	Regulation retrieval and LLM reasoning	Connecting applicable rules and evidence
NL2TL; nl2spec [7, 8]	Formalizing language requirements	Temporal-logic translation; interactive revision	Language-specification correspondence
ACE [9]	Unambiguously interpreted language	Restricted grammar and formal interpretation	Principles of controlled expression
GuardSynth	Structuring constraints for CoC learning	EBLC → logical checks → CNL → CoC augmentation	Linking constraint model, verification, and learning inputs

We assess learning utility by comparing CoC alone (P0) with constraint-augmented CoC (P1 and P2), using the same model and matched training budgets. P1 is our existing natural-language constraint approach, and P2 uses CNL generated from EBLC. Thus, P1/P2 versus P0 is the primary comparison, while P1 versus P2 compares representations within our proposed family. This experiment evaluates the effect of supplying constraints; it does not isolate a causal effect of logical verification. The paper covers

constraint specification, verification, CNL generation, and learning use. EBLC-based test generation and runtime monitoring are subsequent applications.

References

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